

Eukaryotes, Prokaryotes & the Scale of Cells

Quick reference sheet · Biology & Maths · keep handy for revision

1. Eukaryotic vs prokaryotic

Animal and plant cells are **eukaryotic**: genetic material is enclosed in a nucleus. Bacterial cells are **prokaryotic**: much smaller, with a single DNA loop (plus possible plasmids) free in the cytoplasm, not in a nucleus.

Feature	Eukaryotic (animal/plant)	Prokaryotic (bacteria)
Size	Larger	Much smaller
Genetic material	Enclosed in a nucleus	Single DNA loop, free in cytoplasm
Plasmids	None	May have one or more
Cytoplasm & cell membrane	✓	✓
Cell wall	Plants only (cellulose)	✓

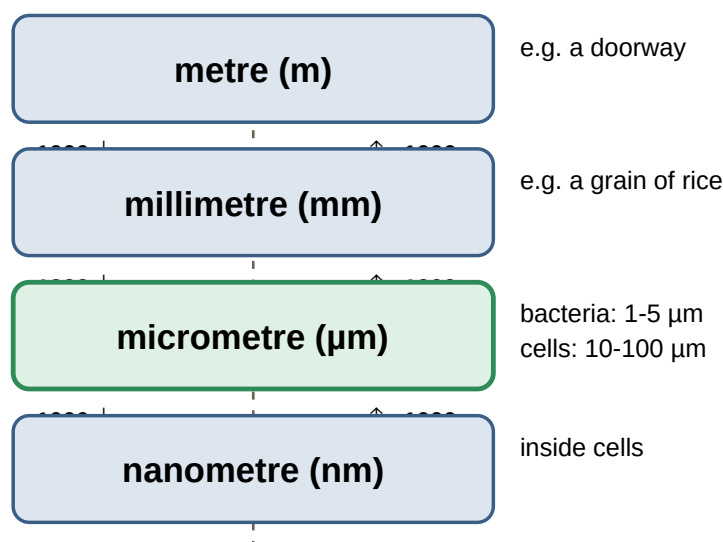
The one thing to remember: nucleus present → eukaryotic. No nucleus (DNA loop instead) → prokaryotic.

Where the names come from: "eukaryotic" = true nucleus; "prokaryotic" = before nucleus. Animal and plant cells seal their DNA inside a nuclear membrane; a bacterium's DNA is a single circular strand loose in the cytoplasm, with no membrane around it.

Plasmids are small circular pieces of DNA, separate from the main DNA loop — much smaller, carrying just a few genes (e.g. antibiotic resistance) rather than the full genome. They can be copied and passed between bacteria independently of the main loop, which is how antibiotic resistance spreads, and they can be extracted and modified to make bacteria produce useful proteins such as insulin (genetic engineering).

2. The scale of cells

Each step down this ladder is ×1000 smaller (each step up is ÷1000):



Bacteria are about 1–5 µm across; animal and plant cells are about 10–100 µm across — roughly 10× bigger.

Worked conversion: $0.02 \text{ mm} \times 1000 = 20 \text{ µm}$ (mm → µm, multiply by 1000; µm → mm, divide by 1000).

You may also see **centimetres (cm)**: $1 \text{ cm} = 10 \text{ mm}$. Cells are never measured in cm — it only appears on the exam skills list.

3. Standard form for tiny numbers

Standard form: $A \times 10^n$, where A is between 1 and 10.

Given	In metres	Standard form
2 µm	0.000002 m	$2 \times 10^{-6} \text{ m}$
25 µm	0.000025 m	$2.5 \times 10^{-5} \text{ m}$

Shortcut: $1 \text{ µm} = 10^{-6} \text{ m}$, so attach $\times 10^{-6}$ to a µm value and adjust so the first digit is between 1 and 10.

4. Order of magnitude calculations

An "order of magnitude" means a **factor of 10**. Divide the bigger size by the smaller size to find how many times bigger it is.

Worked example: bacterium 2 µm, animal cell 20 µm. $20 \div 2 = 10$ — the animal cell is one order of magnitude bigger.